

LISBOA PORTELLA AP, Portugal

WMO: 085360

Lat: 38.781N

Lon: 9.136W

Elev: 374

StdP: 14.50

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.7	42.8	26.8	21.0	50.2	30.4	24.9	51.8	25.2	55.8	22.7	56.0	6.9	50	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.3	93.4	67.1	89.7	66.6	86.2	66.2	71.0	82.8	70.0	80.8	69.0	79.1	10.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	105.4	71.6	67.6	103.0	71.4	66.2	98.0	71.1	35.1	83.2	34.1	81.3	33.4	79.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	21.0	18.9	DB	36.1	101.0	2.7	3.9	34.2	103.7	32.6	106.0	31.1	108.2	29.2	111.0	(4)
(5)			WB	33.2	73.3	2.7	1.1	31.3	74.1	29.7	74.8	28.1	75.4	26.2	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	52.9	54.1	58.1	60.8	65.2	70.7	73.6	75.0	72.1	66.7	58.4	53.9	(6)	
	DBStd	9.15	4.26	4.00	4.74	5.25	5.61	5.56	4.95	4.79	4.33	4.52	4.50	4.42	(7)	
	HDD50	50	21	11	2	0	0	0	0	0	0	0	1	15	(8)	
	HDD65	1705	377	306	221	150	65	8	0	0	0	32	201	345	(9)	
	CDD50	4975	110	124	252	323	471	621	732	775	662	517	254	134	(10)	
	CDD65	1156	0	0	6	23	71	179	267	310	212	84	4	0	(11)	
	CDH74	7689	0	0	39	140	507	1270	1946	2283	1179	323	2	0	(12)	
	CDH80	2790	0	0	1	22	160	491	765	908	377	66	0	0	(13)	
(14) Wind	WSAvg	9.3	8.0	8.7	9.1	9.5	10.0	10.4	11.5	10.8	9.1	8.2	8.1	8.2	(14)	
(15) Precipitation	PrecAvg	28.9	4.1	3.6	2.8	2.4	1.9	0.7	0.1	0.2	1.2	3.7	4.8	4.0	(15)	
	PrecMax	46.8	13.9	11.7	10.5	6.4	6.1	3.9	1.7	2.3	5.1	10.3	14.7	15.1	(16)	
	PrecMin	16.4	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	(17)	
	PrecStd	8.1	3.0	2.6	2.3	1.6	1.6	0.9	0.3	0.4	1.3	2.9	3.4	3.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	68.1	78.6	82.8	89.9	95.3	98.5	98.8	93.0	86.2	73.4	65.9	(19)	
		MCWB	58.9	57.9	59.0	62.7	66.0	68.2	68.3	67.8	66.7	63.7	61.8	61.6	(20)	
	2%	DB	62.7	64.7	73.7	78.4	84.3	89.9	93.1	93.4	88.1	80.8	69.8	64.2	(21)	
		MCWB	58.5	56.6	58.1	61.5	64.2	66.9	67.2	67.2	65.6	63.6	63.7	61.8	(22)	
	5%	DB	60.9	62.6	70.0	73.7	79.2	85.9	87.8	89.3	84.4	77.2	67.7	62.6	(23)	
		MCWB	57.9	56.0	57.3	60.0	63.1	66.1	67.1	66.8	65.4	63.6	62.3	60.1	(24)	
	10%	DB	59.3	60.7	66.1	69.8	75.2	81.0	83.8	84.4	80.6	73.7	65.8	60.9	(25)	
		MCWB	56.3	56.1	56.6	59.2	62.1	65.2	66.5	66.4	65.0	63.3	61.1	58.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	61.2	62.9	66.6	69.9	71.1	72.0	72.7	71.4	70.7	68.2	64.5	(27)	
		MCDB	63.0	63.8	70.5	78.5	83.4	87.3	88.6	86.7	79.7	74.5	70.1	64.8	(28)	
	2%	WB	60.4	59.9	60.9	63.5	67.1	69.4	70.2	71.0	69.9	68.9	66.2	62.7	(29)	
		MCDB	61.6	62.4	66.5	73.6	77.9	83.4	84.4	82.6	77.7	73.2	67.7	63.5	(30)	
	5%	WB	59.0	58.6	59.7	61.7	65.0	67.8	69.1	69.9	68.9	67.5	64.1	60.9	(31)	
		MCDB	60.4	61.1	65.2	70.0	74.7	80.6	81.9	80.5	76.4	71.6	66.1	62.1	(32)	
	10%	WB	57.4	57.1	58.4	60.1	63.5	66.5	67.8	68.7	67.7	66.4	61.9	58.9	(33)	
		MCDB	59.2	59.7	64.0	66.9	72.1	77.6	79.3	78.8	74.9	70.4	64.4	60.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.2	13.8	14.2	15.3	16.9	17.8	18.3	16.4	13.5	11.1	10.4	(35)	
		MCDBR	11.3	14.5	20.4	20.6	22.9	24.2	24.8	24.5	22.5	18.8	12.9	10.2	(36)	
	5% WB	MCWBR	8.1	8.8	9.8	10.1	10.4	9.6	8.9	8.8	8.3	8.0	7.9	7.9	(37)	
		MCWBR	8.7	11.3	15.5	17.5	18.4	20.6	20.3	18.9	16.4	11.8	9.8	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.338	0.367	0.384	0.392	0.397	0.381	0.394	0.393	0.360	0.336	0.326	(40)	
		taud	2.517	2.477	2.406	2.352	2.361	2.367	2.402	2.376	2.391	2.504	2.528	2.516	(41)	
	Ebn at Noon		270	281	283	283	282	279	283	277	270	270	264	261	(42)	
		Edh at Noon	24	29	34	38	39	39	37	37	35	28	24	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		710	1027	1395	1796	2154	2326	2369	2151	1683	1157	790	641	(44)	
	RadStd		89	120	148	147	144	121	94	82	107	95	97	72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		-0.54	N/A	N/A	N/A	N/A	-0.81	N/A	N/A	N/A	N/A
Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page